

Principles of Applied Engineering

TEKS §127.781 • CTE Level 1 • Grade 9 • Mr. Gavaldon • Da Vinci School

Week 2

MONDAY

Day 6 - Build-a-Car: Plan & Design

Aug 17

OBJECTIVE

Students will define the roll-far problem with measurable success criteria and constraints, generate and score two car concepts in a weighted decision matrix, and predict roll distance from an energy model ($PE = mgh$) - completing an engineering planning packet before any building.

TEKS

§127.781(d)(2)(G) engineering design process — planning; (d)(3)(C) sketching the design

ELPS / EB SUPPORT

Design brief and planning packet give sentence frames plus a labeled force diagram (VC, DP, CD). Pre-teach constraint / criteria / rolling-resistance from a picture bank (PV). Mixed teams with builder / recorder / tester roles; native-language partner talk before writing the justification (PN, WT).

AGENDA • 45 MIN

5' Bell-work: list every factor that could make one car roll farther than another

10' Design brief: the 3-day mission, constraints vs. criteria, the decision matrix

12' Concept: energy (PE to KE) and why cars stop - rolling resistance & alignment

6' Worked example: predict a car's roll distance from height

9' You do: generate 2 concepts, score the matrix, predict distance

3' Exit ticket: constraint vs. criterion

SLIDES / ACTIVITY

Engineering a Car That Rolls: Plan & Design

NOTEBOOK

Glue in the car planning sheet. Sketch your design and label the body, axles, and wheels.

TURN IN

Engineering Design Packet - Roll-Far Car

TUESDAY

Day 7 - Build Day

Aug 18

OBJECTIVE

Students will fabricate their prototype car from an approved plan using safe tool practice, achieve a running clearance fit at the wheel-axle joint, and verify axle spacing and parallelism to a stated tolerance (60.0 ± 1.0 mm), logging every measurement and corrective action.

TEKS

§127.781(d)(2)(G) build the prototype (design process); (d)(4) safe, responsible use of tools & materials

ELPS / EB SUPPORT

Teacher models each assembly step at the front and leaves an annotated photo sequence up (VC, DP, RS). Build-log terms (tolerance, clearance, parallel) pre-taught with a picture bank (PV, CD). Roles keep every student hands-on; partner talk before recording corrective action (PN, WT).

AGENDA • 45 MIN

5' Bell-work: name two build risks and the one dimension that must be in tolerance

7' Safety protocol + fit & tolerance mini-lesson

6' Teacher models axle/wheel assembly and how to measure it

22' Build from approved plans: assemble, measure, roll-test as you go

5' Dimensional inspection + park it + log one fix

SLIDES / ACTIVITY

Build Day: Precision, Fit, and Safe Fabrication

NOTEBOOK

Log your build: what went well, what didn't, and the one fix you'll make on test day.

TURN IN

Build Log & Dimensional Inspection

OBJECTIVE

Students will design a fair test by controlling variables, collect multiple-trial distance data, compute mean, range, and percent improvement, visualize before/after results in a bar chart, and justify their next iteration with the data.

TEKS

\$127.781(d)(2)(G) test & evaluate (design process); (d)(3)(D) present results with visualization

ELPS / EB SUPPORT

Data sheet is mostly numbers with one sentence of reasoning (low language load, CD, DP). Pre-teach independent/dependent/controlled variable and mean/range with a picture bank (PV). Pair-share the analysis before writing the conclusion (WT, PN).

AGENDA • 45 MIN

5' Bell-work: what makes a test fair? name the three kinds of variables

8' Fair testing, variables, and why we run multiple trials (mean & range)

6' Worked example: mean, range, and percent improvement

18' Ramp test: 3 baseline trials -> one controlled change -> 3 trials

5' Analyze: compute percent change and graph before/after

3' Exit ticket: which change to make next and why

SLIDES / ACTIVITY

Test, Measure, Improve: Data-Driven Iteration

NOTEBOOK

Record both trial distances and the single change you made. Was your prediction right?

TURN IN

Test Data & Iteration Analysis

OBJECTIVE

Students will explain what CAD and parametric modeling are and why engineers design in software before building, analyze CAD vs. paper prototyping against engineering criteria, apply professional-online-conduct standards, and confirm a working school email for account creation.

TEKS

\$127.781(d)(3)(C) computer-aided design & drafting (CADD); (d)(1)(A) professional online conduct

ELPS / EB SUPPORT

Video has a guided sheet with sentence stems and target terms to catch (RS, CD, DP). Pre-teach CAD / parametric / extrude with labeled screenshots (PV, VC). 'Show me your email' is a low-stakes one-on-one check with wait time (WT); native-language partner support at the station (PN).

AGENDA • 45 MIN

5' Bell-work: what is CAD, and why not just build every idea by hand?

12' What is CAD? Parametric modeling and the sketch-to-feature workflow

10' Watch: first look at Fusion 360 (guided viewing sheet)

8' Why CAD beats paper + the certification path + professional online conduct

7' Email-confirm station + CAD-vs-paper decision matrix

3' Exit ticket: define parametric

SLIDES / ACTIVITY

From Paper to Parametric CAD

NOTEBOOK

Note 3 things Fusion can do, and one thing you want to build in it this year.

TURN IN

CAD Concepts, Guided Viewing & Account Readiness

OBJECTIVE

Students will create an Autodesk education account, launch and navigate the Fusion 360 interface, model a first parametric part (a dimensioned, extruded 60 x 30 x 3 mm chassis plate), save it to the cloud, and support a peer using a structured help protocol.

TEKS

§127.781(d)(3)(C) set up & navigate CAD (CADD); (d)(1)(B) collaborate & support peers

ELPS / EB SUPPORT

Account and modeling steps are numbered and screenshotted in Classroom (VC, DP, CD). Pre-teach interface terms (sketch, extrude, ViewCube, timeline) with labeled images (PV). Finishers pair with peers using an 'ask three, then me' protocol; native-language walkthrough allowed (PN, WT).

AGENDA • 45 MIN

5' Bell-work: why an education license and a school email for this account?

7' Account steps + tour of the Fusion interface (toolbar, browser, timeline, ViewCube)

8' Demo: sketch -> dimension -> extrude a 60 x 30 x 3 mm chassis plate

20' Create/verify account, launch Fusion, model your first part, save to cloud

5' Peer support + first-look checklist + exit

SLIDES / ACTIVITY

Launch Fusion 360: Your First Parametric Part

NOTEBOOK

Paste your Fusion account email (not password). Note where the Sketch button and timeline are.

TURN IN

Fusion 360 First Session - Setup, Build Log & Peer Support